

Evaluating asymptotic intervals of FH estimates with and without variance smoothing, using simulated data mirroring Kenya height-for-age

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The goal of this simulation study is to evaluate how well variance smoothing can improve interval properties under the FH model. We simulate data meant to mirror the height-for-age z-score in the 2022 Kenya DHS.

Simulation Design

1. Obtain weighted stratified mean estimates $(\boldsymbol{\mu}_R, \boldsymbol{\mu}_U)$, at the desired level, using the DHS data (height-for-age)
2. Generate a continuous response, for each sampled individual k in cluster c , in strata h , in area i , from either
 - a. no cluster effect:

$$y_{ihck}^{(s)} \sim \mathcal{N}(\mu_{ih}, \sigma_{ih}^{2(s)})$$

- b. cluster effect:

$$y_{ihck}^{(s)} \sim \mathcal{N}(\mu_{ih} + \epsilon_c^{(s)}, \sigma_{ih}^{2(s)})$$

where $\sigma_{ih}^{2(s)}$ is generated from,

$$\begin{aligned} \log(\sigma_i^{2(s)}) &= \alpha + Z\gamma + v_i \\ \sigma_{iU}^{2(s)} &= (1 + \kappa_i)\sigma_{iR}^{2(s)} \end{aligned}$$

where Z is a covariate matrix, and v can be distributed either IID or BYM2.

3. Take a two-stage stratified sample, with sampling design and sizes chosen to mirror the Kenya 2022 DHS.
4. Fit models using a mean model with BYM2 covariates and the following variance models:
 - a. IID variance and constant strata effect:

$$\begin{aligned} \log(\sigma_{iR}^{2(s)}) &\sim \mathcal{N}(\alpha, 1/\tau) \\ \sigma_{iU}^{2(s)} &= (1 + \kappa)\sigma_{iR}^{2(s)} \end{aligned}$$

- b. Structured variance and constant strata effect:

$$\begin{aligned} \log(\sigma_{iR}^{2(s)}) &\sim \mathcal{N}(\alpha, 1/\tau) \\ \sigma_{iU}^{2(s)} &= (1 + \kappa)\sigma_{iR}^{2(s)} \end{aligned}$$

where Z is a covariate matrix, and v can be distributed either IID or BYM2

c. IID variance and varied strata effect:

$$\log(\sigma_i^{2(s)}) = \alpha + Z\gamma + v_i$$

$$\sigma_{iU}^{2(s)} = (1 + \kappa_i)\sigma_{iR}^{2(s)}$$

d. Structured variance and varied strata effect:

$$\log(\sigma_i^{2(s)}) = \alpha + Z\gamma + v_i$$

$$\sigma_{iU}^{2(s)} = (1 + \kappa_i)\sigma_{iR}^{2(s)}$$

5. Compare each model to the oracle model (where sampling variance estimate is correct), standard Fay-Herriot (no smoothing) and the naive variance smoothing model with $df = \#clusters - \#strata$.